

# Data Science

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## Lecture 01: Introduction

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# Module 1

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## The Data Deluge

**Guiding Question:** *We generate more data than we can store. Does having more of it make us know more?*

# How Much Data Exists?

| Year | Global datasphere             | Per person on earth           |
|------|-------------------------------|-------------------------------|
| 2002 | 5 exabytes of new information | —                             |
| 2010 | ~1 zettabyte                  | a trillion gigabytes in total |
| 2020 | ~40 zettabytes                | ≈ 5,200 GB                    |
| 2025 | ~175 zettabytes               | ≈ 21,000 GB                   |

In 2007 the amount of information produced first **surpassed available storage capacity**. Since then, most data that is generated is never stored.

# Growth Is Not the Interesting Part

## Volume

More bytes than anyone can read, index or curate.

## Heterogeneity

Text, sensors, images, logs, transactions — rarely in a common schema.

## Velocity

Streams that must be acted on before they can be stored.

The scarce resource is not data. It is **the ability to turn data into justified conclusions** — which is what this course is about.

# Science at Scale

## Large Hadron Collider

1 MB per event, 600 million events per second — roughly **18 zettabytes per year** if everything were kept.

A single tier-1 centre stores ~2.2 petabytes per year.

## The filter is the experiment

Almost all events are **discarded in hardware**, in real time, by trigger rules written in advance.

What is kept is decided by a theory of what matters.

# Data at Web Scale

## Search

Web-scale indexing processes petabytes per day and answers in milliseconds.

## Platforms

Interaction logs — clicks, dwell time, scrolls — dwarf the content itself.

## Models

Training corpora for language models are measured in trillions of tokens.

In every case the value comes from **aggregation across many weak signals**, not from any single record.

Data is never raw: at the LHC a physics theory decides what is recorded, and on the web value comes from aggregating signals that are individually worthless. **Someone always chose what the dataset contains.**

# Module 2

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## Does Data Speak for Itself?

**Guiding Question:** *Twenty-nine teams analysed the same dataset and the same question. Why did they not get the same answer?*

# "The End of Theory"

*With enough data, the numbers speak for themselves. Correlation supersedes causation, and science can advance without coherent models or unified theories.*

**Chris Anderson, *Wired*, 2008.** The argument has returned, almost verbatim, in claims that sufficiently large models make domain theory unnecessary.

**Is it right?**

# One Dataset, Many Answers

Twenty-nine research teams received **the same dataset** and the same question:

*Are football referees more likely to give red cards to dark-skinned players than to light-skinned players?*

## The result

Odds ratios ranged from **0.89 to 2.93**.  
Twenty teams found a significant effect  
nine did not.

## The reason

No team made an error. They made  
**different defensible modelling choices**  
— which covariates, which exclusions,  
which model family.

The data did not determine the answer. The **analysis** did. (Silberzahn et al. 2018)

# Correlation Is Not Causation

Spurious correlations are easy to find and get **easier as datasets grow**: with enough variables, some pair will track another closely by chance alone.

## Why it happens

Testing many pairs makes an extreme correlation likely *somewhere*, even when every pair is independent.

## Why more data does not fix it

More rows shrink the confidence interval around a **meaningless** number. The estimate gets more precise, not more valid.

# Summary Statistics Hide the Data

Datasets whose **summary numbers agree exactly** — same averages, same spread, same strength of association — can look completely different when plotted: a dinosaur, a star, a set of lines (Matejka & Fitzmaurice 2017).

We will make "average", "spread" and "association" precise in Lecture 02, and meet this phenomenon again there as Anscombe's quartet.

Plot the data. Then plot it again, differently.

# Where This Leaves Us

## Data does not speak for itself

Every dataset was selected, filtered and encoded by someone.

## Data cannot be read intuitively

Human intuition fails at scale, in high dimensions, and with base rates.

## So theory is needed *more*

Models make the assumptions explicit enough to be criticised.

Big data does not end theory. It **raises the cost of implicit theory**, because assumptions that stay unstated now get automated and scaled.

# Module 3

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## What Is Data Science?

**Guiding Question:** *If data science is not statistics, not machine learning and not engineering, what is left that is its own?*

# A Definition

**Data Science** researches and applies technologies, systems and algorithms to gain knowledge from potentially large, heterogeneous and dynamically changing data sets — for a particular domain. (Dhar 2013)

The phrase doing the work is "**for a particular domain**": without it, there is no criterion for which patterns count as knowledge.

# Neighbouring Fields

## Machine Learning

Learning from past experience to improve at a task in the future.

## Artificial Intelligence

Building systems that behave intelligently.

## Data Engineering

The twin field: managing, organising and providing access to data.

Data science is not a subset of any of them. It is the discipline of getting from a **domain question** to a **defensible answer**, using whichever of these tools fits.

# Three Kinds of Task

## Explorative

Find patterns.

*Which customers behave similarly?*

## Descriptive

Summarise what is there.

*What does an average customer spend?*

## Predictive

Anticipate future outcomes.

*Is this transaction fraudulent?*

Naming the task first constrains everything else — the method, the evaluation measure, and what counts as success.

Data science is defined by its **question and its domain**, not by its tools. Explorative, descriptive and predictive tasks demand different methods and different evidence — so the first decision in any project is which one you are doing.

# Module 4

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## How Knowledge Is Acquired

*Guiding Question: A pattern in data is not yet knowledge. What has to happen in between?*

# Three Modes of Reasoning

|                                       | From              | To                                    | What it guarantees                       |
|---------------------------------------|-------------------|---------------------------------------|------------------------------------------|
| <b>Deduction</b> <i>(logic)</i>       | premises          | what <b>must</b> follow               | certain — but adds no new content        |
| <b>Induction</b> <i>(statistics)</i>  | observed patterns | a <b>regularity</b> , a hypothesis    | nothing — content-adding, not guaranteed |
| <b>Abduction</b> <i>(probability)</i> | observations      | the <b>best available explanation</b> | nothing — only comparable against rivals |

Machine learning lives in the bottom two rows.



# The Scientific Reasoning Cycle

The three modes are not alternatives — they take turns.

**Example: gravity.**

## Deduction

From the theory: all objects of equal mass fall at the same rate everywhere on earth.

## Induction

Design an experiment, measure fall times at several locations, compute means and standard deviations.

## Abduction

Times agree at one location but vary between locations.  
**Best explanation:** distance from earth's mass centre.

The cycle turns: the abducted explanation becomes the next theory to deduce from. A data-science project runs the same loop, usually several times.

# Mini-Exercise

A dashboard shows that customers who use the mobile app churn **less** than those who do not.

- 1 What would a purely **inductive** reading conclude?
- 2 Give two different **abductive** explanations.
- 3 What **deduction** would let you distinguish them?

# Mini-Exercise

A dashboard shows that customers who use the mobile app churn **less** than those who do not.

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## INDUCTION

"App use reduces churn" — a regularity read straight off the data.

## ABDUCTION

The app causes loyalty — *or* loyal customers install the app. Same correlation, opposite causal direction.

## DEDUCTION

If the app causes loyalty, randomly prompting a subgroup to install it should lower their churn — an experiment deduced from the hypothesis.

# What Counts as Knowledge?

**Knowledge discovery in databases (KDD)** is the non-trivial process of identifying patterns in data that are

- **valid** — they hold beyond the sample,
- **novel** — they were not already known,
- **relevant** — they matter for a decision,
- **interpretable** — they can be understood and acted upon.

A pattern that fails **any** of the four is not knowledge. Most patterns a large dataset offers fail at least one.

≡ Knowledge is a pattern that is **valid, novel, relevant and interpretable**. Data mining produces patterns only the four criteria turn one into knowledge — and most patterns fail at

least one of them.

# Module 5

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## The Process and the Craft

**Guiding Question:** *Most of a data-science project is spent before any model is fitted. On what, and why does it matter?*

# Step 1 — Selection of Data

## Types of source

- **Unstructured:** flat files, documents, text and multimedia
- **Structured:** relational databases, XML, web APIs

## Types of data

media · object-relational · spatial and spatio-temporal · graphs and networks · time series and sequences · data streams

Selection is where the **first and largest** bias enters. What was not selected cannot be discovered later.

# Step 2 — Preprocessing

## Selected

manually or  
automatically, e.g. by  
crawling

## Cleaned

errors and noise  
removed — *garbage  
in, garbage out*

## Integrated

combined with other  
sources, resolving  
conflicts

## Transformed

into a reusable  
representation for  
analysis

This step routinely consumes most of a project's effort — and every choice in it is a modelling decision, not a clerical one.

# Steps 3–5 — Mining, Evaluation, Deployment

## Data mining

Apply the algorithm — clustering, classification, rule mining — that fits the task.

## Evaluation

Test the patterns against the four criteria: valid, novel, relevant, interpretable.

## Deployment

Put the result where a decision is made, and monitor it there.

The process is a **loop**, not a pipeline. Evaluation routinely sends you back to selection.

# What the Work Actually Looks Like

## Ask

Turn a domain question into something a dataset could answer — and check that it is the question worth answering.

## Build

Acquire, clean, model, evaluate. Iterate, mostly backwards.

## Communicate

Deliver a result someone can act on, with its uncertainty attached.

# Three Skill Areas

## Domain knowledge

What the data means, which values are impossible, which differences matter.

## Statistics and modelling

What may be inferred, with what uncertainty, under which assumptions.

## Engineering

Getting the data, at scale, reproducibly, into production.

The interesting failures happen **between** the three: a technically flawless model of the wrong quantity, or a correct analysis nobody can deploy.

The process is a **loop, not a pipeline**, and it spans three skill areas that nobody holds equally. The characteristic failures happen at the seams — a flawless model of the wrong quantity, or a correct analysis nobody can deploy.

# Module 6

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## Applications, and Their Failures

***Guiding Question:*** *What happens to a model's validity once the world can see the model?*

# Web and Social Media

## Google Flu Trends

Search queries were used to nowcast influenza faster than public health reporting.

## And then it failed

It overestimated cases substantially, because search behaviour changed — partly *because of media coverage of the tool itself*.

A model that acts on the world **changes the world it measures**. Validity is not a property a model keeps once earned.

# Text and Language

## Classic tasks

Summarisation, entity extraction, sentiment analysis, question answering — each once a research programme of its own.

## Today

A single pretrained model handles all of them, with the task specified in words rather than in code.

What did **not** change: the need to evaluate on data the model has not seen, and to state what the output may be used for.

# Business and the Public Sector

Reported potentials from the era of the first big-data wave: large retail revenue gains, hundreds of billions in healthcare savings, comparable figures for e-government.

These are **projections from consultancy reports**, not measured outcomes. Treat them as claims to be interrogated — whose baseline, whose costs, and what happened afterwards?

Applications show what is possible their failures show what must be checked. Ask of every deployed model: what changed in the world because it exists?

# Image Credits & References

Dhar, Vasant (2013). Data science and prediction. *Communications of the ACM*.

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